

Background

This tool has six categories of performance, and each are weighted when calculating an overall ‘utility’ for the package. It is possible to change the weightings, but there is no error handling so you must ensure they make sense yourself 😊 .

Table 1 - Category weighting

Performance Category	Default weighting (percent)
Crop value	50
Direct costs	12.5
Environmental impact	
<i>Non-pesticide impacts</i>	6.25
<i>Pesticide impacts</i>	6.25
Human health	
<i>Non-pesticide impacts</i>	6.25
<i>Pesticide impacts</i>	6.25
Time and management	6.25
Third party coordination requirements	6.25

Table 2 – User provided ratings

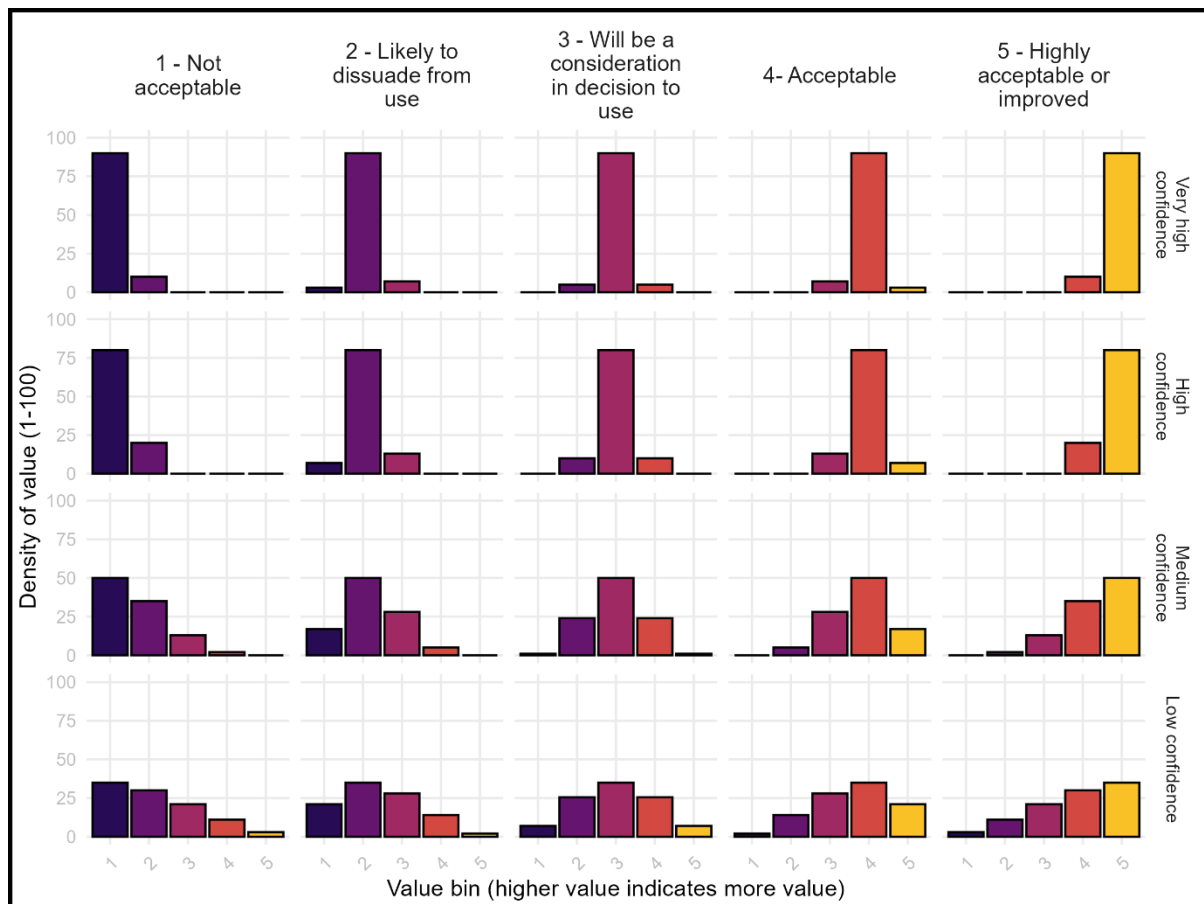
User Rating	Assigned Value
1 – Not acceptable	1
2 – Likely to dissuade from use	2
3 – Will be a consideration in decision to use	3
4 – Acceptable	4
5 – Highly acceptable or improved	5

Note that you will need to assign a confidence to your assessments (Table 3).

Table 3 – User provided confidences

Confidence rating	Percent of distribution allotted to selected rating
Very high	90%
High	80%
Medium	50%
Low	35%

Figure 1 shows how the user rating and confidence scores are combined to create the 'value distribution'.



Questionnaire

The following questionnaire should be filled out for both Package #1 and #2.

Crop value

This is a measure of the success and therefore the acceptability of the yields and quality obtained using this technology/package. The higher the value (in quantity and/or quality) the greater the acceptability.

Crop yield and/or quality when using the technology/package is:

1 – Not acceptable	2 – Likely to dissuade from use	3 – Will be a consideration in decision to use	4 - Acceptable	5 – Highly acceptable or improved
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My confidence in this rating is:

Low	Medium	High	Very high
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Some things to consider:

- Opportunity costs when implementing the practice(s), for example if you take strips of land out of production to install natural habitat to support natural enemies.
- An increase in crop value due to new markets available, for example by producing pesticide-free products

Direct costs

This is a measure of the value delivered against the cost of the package.

The net cost of the technology/package to implement is:

1 – Not acceptable	2 – Likely to dissuade from use	3 – Will be a consideration in decision to use	4 - Acceptable	5 – Highly acceptable or improved
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My confidence in this rating is:

Low	Medium	High	Very high
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Some things to consider:

- Does it require upfront costs that take time to see a return on investment, for example buying a mechanical weeding implement.

- Does it require time before the return on investment is realized, or is it realized immediately, for example planting of a diverse flower buffer versus re-establishing native habitat.

Environmental impact

Non-pesticide impacts

This is for factors **other** than any pesticides used.

The potential environmental impact of the practice(s) is/are:

1 – Not acceptable	2 – Likely to dissuade from use	3 – Will be a consideration in decision to use	4 - Acceptable	5 – Highly acceptable or improved
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My confidence in this rating is:

Low	Medium	High	Very high
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Some things to consider:

- Does the practice impact greenhouse gas emissions, for example by requiring more tractor passes
- Does the practice impact water quality, for example by reducing nitrogen applications
- Does the practice support biodiversity, for example by establishing flower strips along field borders

Pesticide impacts

This reflects the impacts of the pesticides used. You can use the Pesticide Impact Explorer (PIE) Tool to help you estimate these: <https://agro.au.dk/pietool>

The potential environmental impact of the pesticide(s) is/are:

1 – Not acceptable	2 – Likely to dissuade from use	3 – Will be a consideration in decision to use	4 - Acceptable	5 – Highly acceptable or improved
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My confidence in this rating is:

Low	Medium	High	Very high
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Some things to consider:

- What are the societal costs in the environmental fate and/or the ecotoxicology compartments?

Human health and safety

Non-pesticide impacts

This is for factors **other** than any pesticides used, and mainly reflect non-pesticide risks to the workers.

Worker exposure to non-pesticide health and/or safety risks is:

1 – Not acceptable	2 – Likely to dissuade from use	3 – Will be a consideration in decision to use	4 - Acceptable	5 – Highly acceptable or improved
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My confidence in this rating is:

Low	Medium	High	Very high
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Some things to consider:

- Exposure to physical risks, such as the use of ladders when implementing the practice
- Does it involve activities that must be done at night, or during the heat of the day

Pesticide impacts

This reflects the impacts of the pesticides used. You can use the Pesticide Impact Explorer (PIE) Tool to help you estimate these: <https://agro.au.dk/pietool>

Human health exposure to pesticide-based health risks is:

1 – Not acceptable	2 – Likely to dissuade from use	3 – Will be a consideration in decision to use	4 - Acceptable	5 – Highly acceptable or improved
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My confidence in this rating is:

Low	Medium	High	Very high
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Some things to consider:

- What are the combined societal costs in the human health compartment for all pesticides applied?

Time and management

The impact of implementing the package on time and management considerations, outside of those captured by direct costs.

The time and/or effort requirement of the user to implement and/or maintain the effectiveness of the practice(s) is:

Questionnaire to support PPAT assessment of IPM package impacts

1 – Not acceptable	2 – Likely to dissuade from use	3 – Will be a consideration in decision to use	4 - Acceptable	5 – Highly acceptable or improved
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My confidence in this rating is:

Low	Medium	High	Very high
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Some things to consider:

- Does the practice require certain skills or training?
- Does the practice require careful timing of application?

Third party coordination requirements

Whether implementation requires additional supply chains, infrastructure, or cooperation that may be outside of the user's direct control.

The coordination required for the technology/package to be effective is:

1 – Not acceptable	2 – Likely to dissuade from use	3 – Will be a consideration in decision to use	4 - Acceptable	5 – Highly acceptable or improved
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My confidence in this rating is:

Low	Medium	High	Very high
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Some things to consider:

- Can a producer implement the practice(s) immediately, or do they require training to run a spot sprayer
- Can the producer implement it now, or do they need to hire a third-party to, for example, spray pesticides using a drone
- Does the practice require establishment of supply chains, for example a new biological control agent or a new crop variety
- Does the practice require coordination with neighbors, for example establishment of buffer areas